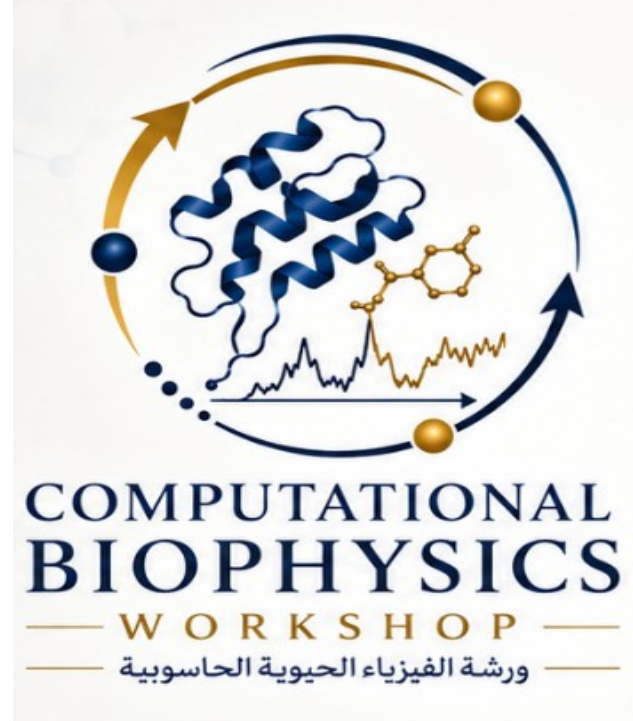


Modeling Small Molecules Using **Avogadro** & **OpenBabel**



Today's **roadmap**

Five short stops — from sketching a molecule on paper to building its 3D shape inside a computer.



01



Why model on computers?



02



Representing molecules

Formulas, skeletal drawings, SMILES strings — what each one tells the computer.



03



Energy landscape & minimization

Why molecules pick certain shapes, and how software finds them.



04



3D file formats

MOL, SDF, XYZ, PDB — the storage formats that carry 3D coordinates.



05



Tools & databases

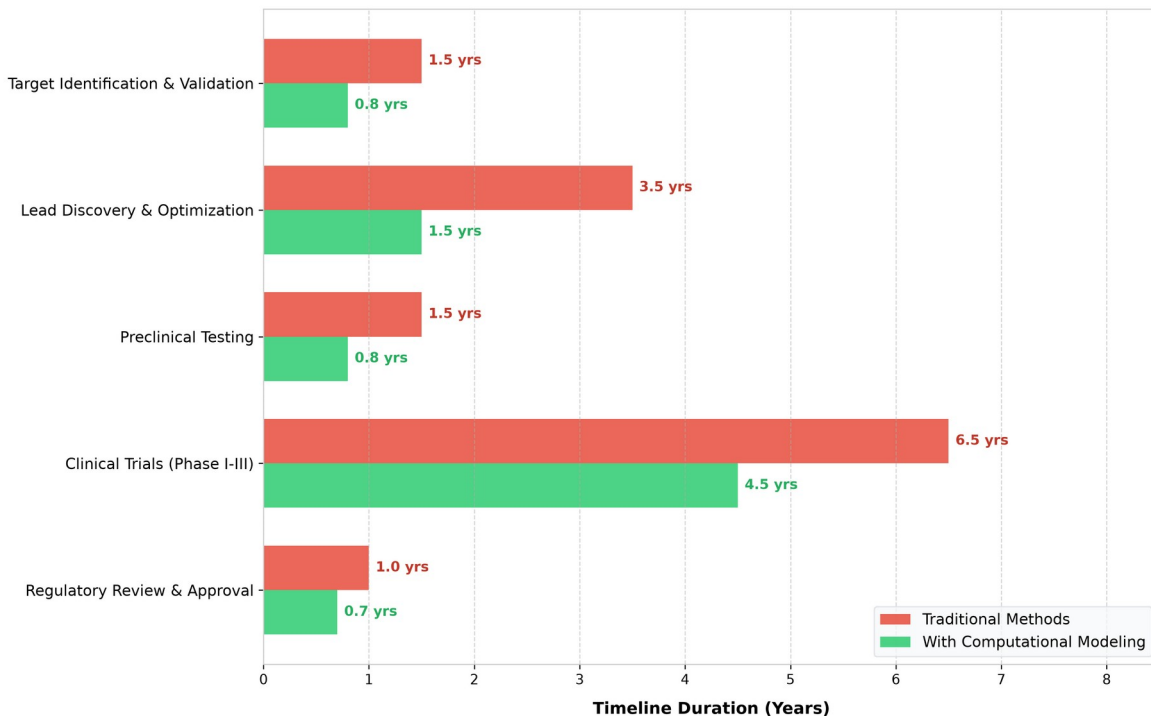
Avogadro, OpenBabel, and where to download ready-made molecules.

00

**Why Model Chemical
Compounds on Computers?
OR
How to treat a new disease?**

Why model chemical compounds on computers? OR How to treat a new disease?

Drug Discovery Timeline & Cost Comparison: Traditional vs. Computational Modeling



ECONOMIC & FINANCIAL IMPACT SUMMARY

- Traditional Total Estimated Cost: \$2.60 Billion | Total Timeline: 14.0 Years
- Computational Modeling Total Cost: \$1.48 Billion | Total Timeline: 8.3 Years
- Total Cost Reduction: \$1.12 Billion (43% savings) | Time Saved: 5.7 Years (41% faster)

01

Representing Molecules Digitally

From paper sketches to text strings that a computer can read, store, and search.

How much does each representation tell us?

The same molecule — ethanol — shown three ways. Each step adds information the previous one lacked.

01 Molecular formula



TELLS YOU

- Atom count
- Elements present

MISSES

- Bond types
- Connectivity
- XYZ Coordinates

02 Condensed structural



TELLS YOU

- Bond types
- Some connectivity
- Functional groups

MISSES

- Connectivity (Branches)
- XYZ Coordinates

03 Skeletal (line)



TELLS YOU

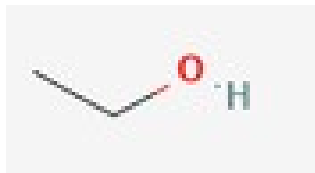
- All connectivity
- Bond type
- Ring info

MISSES

- XYZ Coordinates

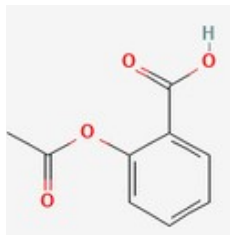
Skeletal formula — the chemist's shorthand

Lines = carbon-carbon bonds, vertices = carbons, and every heteroatom (O, N, S, ...) is drawn explicitly.
Hydrogens on carbons are implicit.



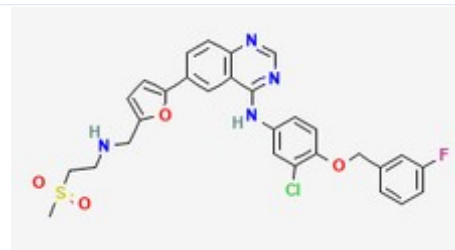
A

Example A — simple hydrocarbon



B

Example B — heteroatoms shown



C

Example C — ring system

WHY IT WORKS

- Captures all connectivity in a compact drawing
- Quick to read for trained chemists
- Standardized internationally (IUPAC)

WHY IT IS NOT ENOUGH FOR A COMPUTER

- Stored as pixels — software cannot easily parse atoms and bonds
- Coordinates vary between drawings of the same molecule
- This is a problem, right?

The answer is simple:



SMILES

What are SMILES?

A single line of text that completely describes a molecule's connectivity — small enough to send in a chat message.

What it stands for

Simplified Molecular-Input Line-Entry System

SMILES encodes **atoms**, **bonds**, **rings**, and **branches** as a string of characters. A program reading the string can rebuild the molecule's graph and even generate its 3D coordinates.

A closely related format is **InChI** (International Chemical Identifier). InChI is more sophisticated and uniquely identifies a structure, but SMILES is the de-facto standard for everyday exchange.

Atoms: **C**, **N**, **O**, ...

Bonds: **-**, **=**, **#**

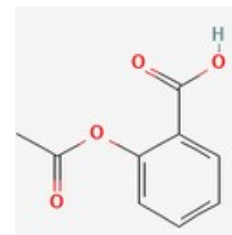
Branches: **()**

ETHANOL IN SMILES

CCO

ASPIRIN IN SMILES

CC(=O)Oc1ccccc1C(=O)O



⚡ Same molecule → always the same string. Tiny, portable, searchable.

SMILES — a quick reference

Six common molecules and their SMILES strings. Notice how rings and branches use parentheses and digits.

Molecule	Formula	SMILES	What the string tells you
Water	H ₂ O	O	Single oxygen, two hydrogens implicit.
Ethanol	C ₂ H ₆ O	CCO	Two carbons then oxygen; H's are implicit.
Benzene	C ₆ H ₆	c1ccccc1	Lowercase 'c' = aromatic carbon; '1' opens & closes the ring.
Aspirin	C ₉ H ₈ O ₄	CC(=O)Oc1ccccc1C(=O)O	Branches in (); aromatic ring with 'c1...c1'.
Caffeine	C ₈ H ₁₀ N ₄ O ₂	CN1C=NC2=C1C(=O)N(C(=O)N2C)C	Two fused rings, multiple branches.
Glucose	C ₆ H ₁₂ O ₆	OCC1OC(O)C(O)C(O)C1O	Linear chain plus a 6-membered ring.

01 Uppercase **C, N, O** = aliphatic atoms; lowercase **c, n** = aromatic.

02 Digits **1–9** pair up to mark ring openings and closures.

03 Branches go inside parentheses.

02

Energy Landscape & Minimization

Why a molecule picks one 3D shape out of many — and how a computer searches for it.

How to design a new molecule?

A metaphor: imagine carving a sculpture. You start with a block of stone (the periodic table) and remove everything that is not the face you want.

◆ THE MODELLING PROBLEM

You know the chemical formula and which atom connects to which, but you do not know where every atom sits in 3D space.

Going from a **SMILES** string (just connectivity) to a **real 3D shape** (coordinates) is the modelling problem we need to solve.

✓ Three ingredients you need

- 01 **Building units** — the atoms and their identities (which element, how many)
- 02 **Connectivity** — single / double / triple bonds, branches, rings
- 03 **A rule for shape** — how atoms interact each other in 3D (the energy landscape)

→ Going from SMILES to a 3D structure requires us to find the molecule's most stable shape — which means we first need to understand the energy landscape.

Energy landscape — the concept

Imagine a hilly surface where height = energy. Every point on the surface is one possible 3D shape (conformation) of the molecule.

WHAT IT IS

The energy landscape is a **multi-dimensional surface** where the potential energy of a molecule is plotted against its geometric parameters — bond lengths, bond angles, and dihedral angles.

WHY IT MATTERS

- **Valleys (minima)** = stable shapes the molecule likes to sit in
- **Peaks (maxima)** = unstable shapes the molecule avoids
- **The lowest valley** = the most stable shape, the one you would actually observe

KEY TERM

Conformation: one specific 3D arrangement of the atoms.

Open the `Energy_landscape_final.html` file



Energy minimization & force fields

How software like Avogadro actually finds the most stable 3D shape of a molecule.

⌘ Energy minimization in 4 steps

1 Initial guess

The software builds a rough 3D structure from the SMILES/formula



2 Compute energy

Calculate the total potential energy using a force field



3 Compute slope (gradient)

Find the direction of steepest energy decrease



4 Adjust & repeat

Move atoms slightly downhill, recompute, until energy stops dropping

The final shape is the lowest-energy conformation the algorithm could find — usually a local minimum, hopefully the global one.

♦ How to calculate energy?

A **force field** is the set of mathematical equations that give you the energy for any given 3D arrangement.

Without a force field, the energy landscape cannot be drawn at all.

♦ Bond stretching

→ Angle bending

↻ Torsional rotation

♦ Non-bonded interactions (van der Waals + electrostatic)

ff = MMFF94 | UFF | GAFF — common force fields used by OpenBabel and Avogadro

CHAPTER

03

3D File Formats

How atoms in 3D space get stored on disk — and which format fits which job.

Different 3D file formats — at a glance

All four formats store atomic coordinates, but each was born for a different purpose. Pick the one that matches your task.



MOL

Connectivity table + 2D or 3D coordinates for a single molecule.

Best for: single small molecule, editable in Avogadro.



SDF

Many MOL records bundled together with extra data fields (charges, logP, etc.).

Best for: molecule libraries, virtual screening.



XYZ

Bare-minimum: just element + x, y, z per atom. No connectivity.

Best for: quantum-chemistry jobs



PDB

Designed for proteins and nucleic acids, but also stores small molecules.

Best for: biomolecules, docking, drug-protein complexes.

◆ **Rule of thumb:** small organic molecule → MOL or SDF
pure coordinates → XYZ
protein or docking → PDB

MOL format

A single molecule with a connectivity table — atoms, bonds, and 3D coordinates in one plain-text block.

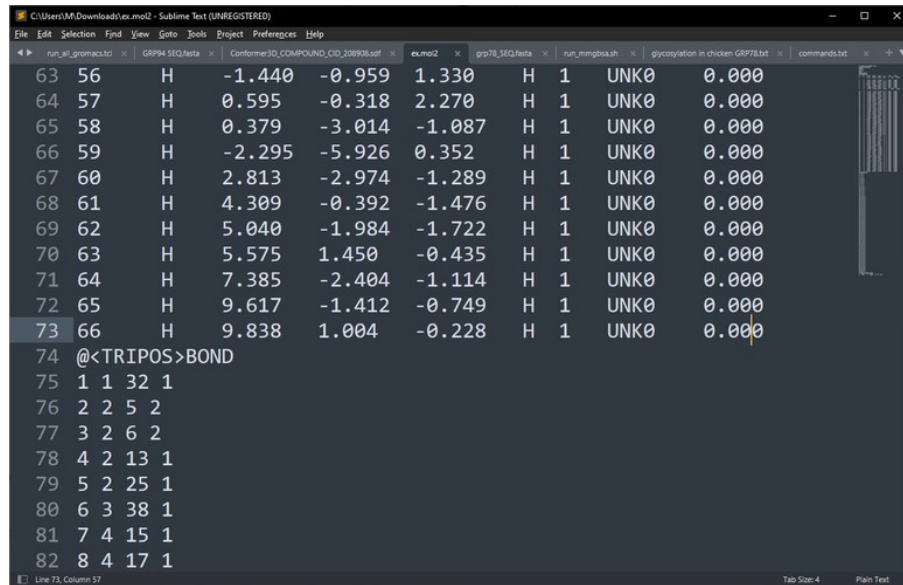
STRUCTURE

- **Header:** molecule name + metadata (3 lines)
- **Atom block:** element + x, y, z for every atom
- **Bond block:** which atoms connect and the bond type

WHEN TO USE

- A single molecule you want to draw or edit in Avogadro
- Sharing one structure with a colleague
- Input to quantum-chemistry or docking programs

① Plain text — open it in any text editor to see the structure inside.



```
C:\Users\MI\Downloads\ex.mol2 - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
run_all_gromacs.txt x GRP94_SEQ.fasta x Conformer3D_COMPOUND_CID_208908.txt x ex.mol2 x grp78_SEQ.fasta x run_mmgbssa.sh x glycosylation_in_chicken_GRP78.txt x commands.txt x
63 56 H -1.440 -0.959 1.330 H 1 UNK0 0.000
64 57 H 0.595 -0.318 2.270 H 1 UNK0 0.000
65 58 H 0.379 -3.014 -1.087 H 1 UNK0 0.000
66 59 H -2.295 -5.926 0.352 H 1 UNK0 0.000
67 60 H 2.813 -2.974 -1.289 H 1 UNK0 0.000
68 61 H 4.309 -0.392 -1.476 H 1 UNK0 0.000
69 62 H 5.040 -1.984 -1.722 H 1 UNK0 0.000
70 63 H 5.575 1.450 -0.435 H 1 UNK0 0.000
71 64 H 7.385 -2.404 -1.114 H 1 UNK0 0.000
72 65 H 9.617 -1.412 -0.749 H 1 UNK0 0.000
73 66 H 9.838 1.004 -0.228 H 1 UNK0 0.000
74 @<TRIPOS>BOND
75 1 1 32 1
76 2 2 5 2
77 3 2 6 2
78 4 2 13 1
79 5 2 25 1
80 6 3 38 1
81 7 4 15 1
82 8 4 17 1
Line 73, Column 57 Tab Size: 4 Plain Text
```

SDF format

A library in a single file — many MOL records stacked together, each followed by tagged data fields like name, logP, or charges.

WHAT IT IS

An SDF (Structure-Data File) is essentially many MOL records concatenated. After each molecule, you can attach extra fields — for example, the compound's name, calculated logP, number of rotatable bonds, or partial charges.

WHY USE IT

- Distribute thousands of molecules in one file
- Carry computed properties alongside each structure
- The standard input format for virtual screening pipelines

```

62      0.3792  -3.0141  -1.0872 H   0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
63     -2.2952  -5.9262   0.3520 H   0  0  0  0  0  0  0  0  0  0  0  0  0  0
64      2.8134  -2.9744  -1.2886 H   0  0  0  0  0  0  0  0  0  0  0  0  0  0
65      4.3092  -0.3916  -1.4764 H   0  0  0  0  0  0  0  0  0  0  0  0  0  0
66      5.0403  -1.9841  -1.7215 H   0  0  0  0  0  0  0  0  0  0  0  0  0  0
67      5.5754   1.4501  -0.4350 H   0  0  0  0  0  0  0  0  0  0  0  0  0  0
68      7.3849  -2.4037  -1.1141 H   0  0  0  0  0  0  0  0  0  0  0  0  0  0
69      9.6174  -1.4119  -0.7494 H   0  0  0  0  0  0  0  0  0  0  0  0  0  0
70      9.8380   1.0038  -0.2283 H   0  0  0  0  0  0  0  0  0  0  0  0  0  0
71      1 32  1  0  0  0  0
72      2  5  2  0  0  0  0
73      2  6  2  0  0  0  0
74      2 13  1  0  0  0  0
75      2 25  1  0  0  0  0

```

```

139 39 65 1 0 0 0 0
140 40 66 1 0 0 0 0
141 M END
142 > <PUBCHEM_COMPOUND_CID>
143 208908
144
145 > <PUBCHEM_CONFORMER_RMSD>
146 1.4
147
148 > <PUBCHEM_EFFECTIVE_ROTOR_COUNT>
149 11
150
151 > <PUBCHEM_HEAVY_ATOM_COUNT>
152 40
153
154 > <PUBCHEM_ATOM_DEF_STEREO_COUNT>
155 0

```

XYZ format

The simplest possible 3D format — just an atom count, a comment line, then one row per atom with element and x, y, z.

STRUCTURE

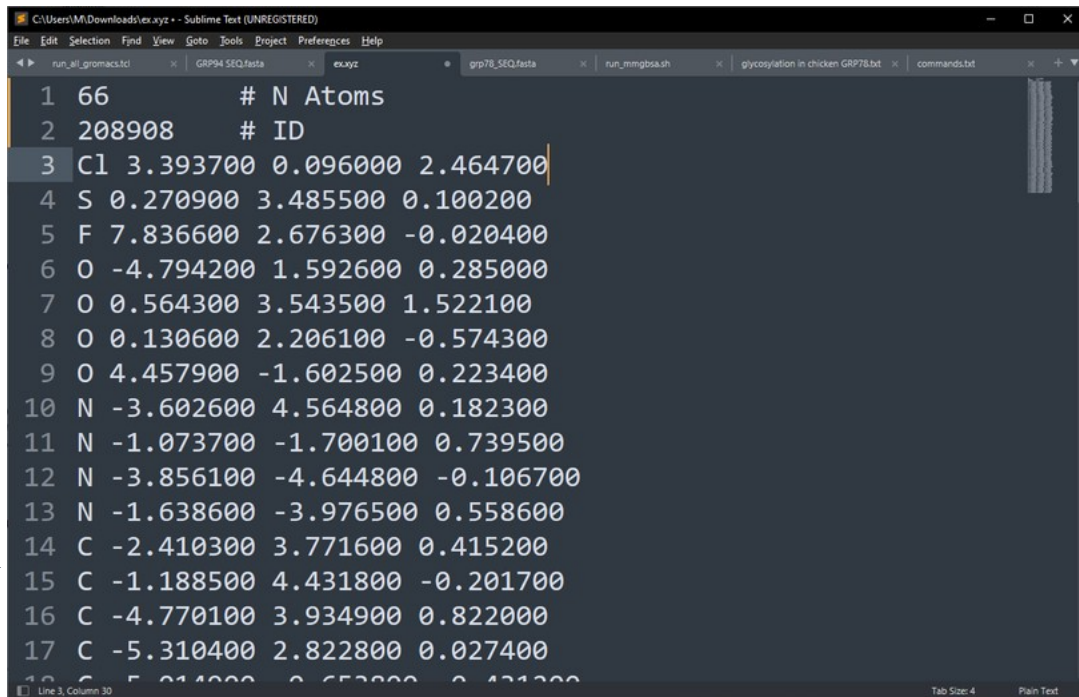
- **Line 1:** number of atoms (an integer)
- **Line 2:** a free-text comment (often the molecule name)
- **Remaining lines:** element + x y z coordinates

PROS

- Trivial to read and write
- No bond information needed
- Standard input for many quantum-chemistry programs (Gaussian, ORCA, etc.)

CONS

- Connectivity is implicit — must be re-derived from distances
- No standard way to carry extra properties



The screenshot shows a Sublime Text editor window with the file 'ex.xyz' open. The file content is as follows:

```
1 66          # N Atoms
2 208908      # ID
3 C1 3.393700 0.096000 2.464700
4 S 0.270900 3.485500 0.100200
5 F 7.836600 2.676300 -0.020400
6 O -4.794200 1.592600 0.285000
7 O 0.564300 3.543500 1.522100
8 O 0.130600 2.206100 -0.574300
9 O 4.457900 -1.602500 0.223400
10 N -3.602600 4.564800 0.182300
11 N -1.073700 -1.700100 0.739500
12 N -3.856100 -4.644800 -0.106700
13 N -1.638600 -3.976500 0.558600
14 C -2.410300 3.771600 0.415200
15 C -1.188500 4.431800 -0.201700
16 C -4.770100 3.934900 0.822000
17 C -5.310400 2.822800 0.027400
```

The editor interface includes a menu bar (File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, Help) and a tab bar at the top showing several open files. The status bar at the bottom indicates 'Line 3, Column 30' and 'Tab Size: 4 Plain Text'.

PDB format

Built for proteins and nucleic acids — the workhorse format of structural biology, also used for small molecules bound to a protein.

STRUCTURE

- Each atom gets a record line starting with **ATOM** or **HETATM**
- Records are grouped into residues (amino acids, nucleotides, ligands)
- Header lines describe the experiment (X-ray resolution, etc.)

WHEN TO USE

- Protein structures from the Protein Data Bank
- Docking a small molecule inside an enzyme
- Visualizing a drug-target complex in PyMOL, ChimeraX, or Avogadro

```

599 ATOM      1  N   ARG A  6      4.165  29.452 -8.677  1.00127.94      N
600 ATOM      2  CA  ARG A  6      3.943  29.214 -7.253  1.00123.31      C
601 ATOM      3  C   ARG A  6      4.037  30.514 -6.450  1.00133.97      C
602 ATOM      4  O   ARG A  6      5.125  30.896 -6.009  1.00144.15      O
603 ATOM      5  CB  ARG A  6      4.955  28.195 -6.718  1.00108.69      C
604 ATOM      6  CG  ARG A  6      4.630  26.733 -6.992  1.00106.31      C
605 ATOM      7  CD  ARG A  6      4.669  26.448 -8.478  1.00115.01      C
606 ATOM      8  NE  ARG A  6      4.174  25.117 -8.811  1.00126.38      N
607 ATOM      9  CZ  ARG A  6      4.914  24.018 -8.786  1.00129.29      C
608 ATOM     10  NH1 ARG A  6      6.183  24.050 -8.416  1.00132.15      N
609 ATOM     11  NH2 ARG A  6      4.364  22.855 -9.129  1.00128.98      N
610 ATOM     12  N   SER A  7      2.899  31.193 -6.271  1.00148.91      N
611 ATOM     13  CA  SER A  7      2.831  32.397 -5.446  1.00149.19      C
612 ATOM     14  C   SER A  7      1.385  32.746 -5.128  1.00134.27      C
613 ATOM     15  O   SER A  7      0.831  33.692 -5.694  1.00113.24      O
614 ATOM     16  CB  SER A  7      3.504  33.587 -6.136  1.00161.26      C
615 ATOM     17  OG  SER A  7      4.829  33.284 -6.511  1.00169.22      O
616 ATOM     18  N   SER A  8      0.778  32.087 -4.114  1.00106.11      N
617 ATOM     19  CA  SER A  8     -0.295  32.740 -3.350  1.00101.20      C
618 ATOM     20  C   SER A  8     -0.965  31.857 -2.296  1.00  97.86      C
619 ATOM     21  O   SER A  8     -0.377  31.609 -1.235  1.00  98.37      O
620 ATOM     22  CB  SER A  8     -1.384  33.305 -4.264  1.00  98.58      C
  
```

Which file format should I use?

There is no single best format — pick the one that matches your task and your downstream tool.

	MOL	SDF	XYZ	PDB
Single molecule?	Yes	Yes (one of many)	Yes	Yes
Many molecules / library?	No	Yes	Trajectory only	Yes (one model)
Stores connectivity?	Yes	Yes	No (implicit)	Yes
Stores extra data fields?	No	Yes	No	Limited (header)
Typical user	Avogadro, sketcher	Virtual screening	Quantum chemistry	Structural biology
Built for	A small molecule	A compound library	Bare coordinates	Proteins & complexes



Sketching or editing one molecule → **MOL**



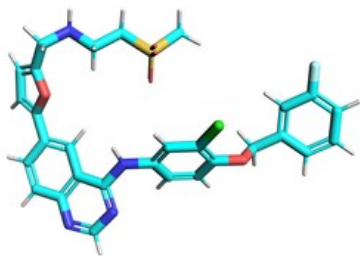
Screening a library of thousands → **SDF**



Docking with a protein target → **PDB**

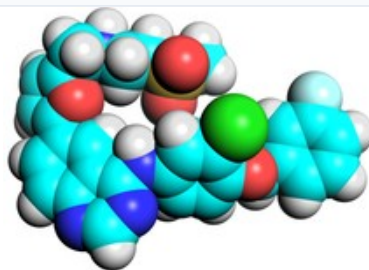
Different 3D visual representations

Once atoms have 3D coordinates, the same molecule can be drawn in several styles — each one tells a different story.



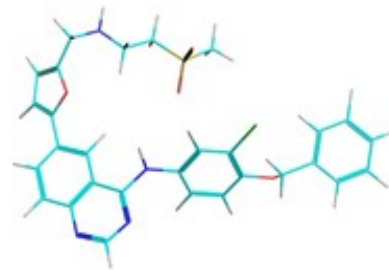
1 Ball and Stick

Spheres represent atoms, sticks represent bonds. Use it to see connectivity, bond angles, and bond orders at a glance. The default view in most molecular editors.



2 CPK / Space-filling

Each atom is drawn as a sphere scaled to its van der Waals radius. Shows the overall molecular shape and surface steric crowding. Often used when discussing steric clashes.



3 Wireframe

Only bonds are drawn as thin lines, atoms are implicit at the vertices. Useful for large molecules where ball-and-stick becomes cluttered.

♦ **Bonus:** surface representations (electrostatic potential, hydrophobicity) are added on top of these base styles to map properties onto the molecular shape.

Small-molecule databases

You do not have to draw every molecule from scratch — these public databases contain millions of ready-made structures you can download for free.



ZINC20

Purchasable compounds for virtual screening

<https://zinc.docking.org/>



ZINC22

Next-generation 3D library for docking

<https://cartblanche22.docking.org/tranches/3d>



ChEMBL

Bioactive molecules with drug-like properties

<https://chembl.gitbook.io/chembl-interface-documentation/downloads>



DrugBank

Approved drugs + drug-target relationships

<https://go.drugbank.com/>



NCI

National Cancer Institute compound sets

<https://wiki.nci.nih.gov/spaces/NCIDTPdata/pages/160989212/Compound+Sets>



COCONUT

COLlection of Open NatUral productS

<https://coconut.naturalproducts.net/>



McuLe

Company database

<https://mcule.com/>



TIP

Most of these databases let you download compounds in **SDF format**.

Programs: Avogadro & OpenBabel

Two free, open-source tools cover the full workflow — from drawing a molecule to running energy minimization in bulk.



Avogadro

A visual editor for 3D molecular structures

- Draw or import a molecule (**SMILES, MOL, SDF, PDB**)
- Optimize geometry with a few clicks (**MMFF94, UFF, ...**)
- Measure bonds, angles, surfaces, and export to many formats

<https://avogadro.cc/>

GUI · BEGINNER-FRIENDLY

♦ OpenBabel

A command-line Swiss-army knife for chemical files

- Convert between **100+ chemical file formats**
- Generate **3D coordinates** from SMILES
- Run energy minimization **in batch** on thousands of molecules

<https://github.com/openbabel/openbabel/releases/download/openbabel-3-1-1/OpenBabel-3.1.1.exe>

CLI · SCRIPTABLE

♦ COMMON OPENBABEL COMMANDS

SMILES → 3D SDF with minimization

```
obabel *.smiles -O comp_*.sdf --gen3D -h --minimize --ff UFF --steps 100000
```

Minimize an existing SDF and convert to PDB

```
obminimize -n 10000 -sd -ff GAFF mol.sdf > mol.pdb
```

Colab notebook: https://colab.research.google.com/drive/1FJcJTD-YW8Fi8TBZR901_PF1PVgmzOWE